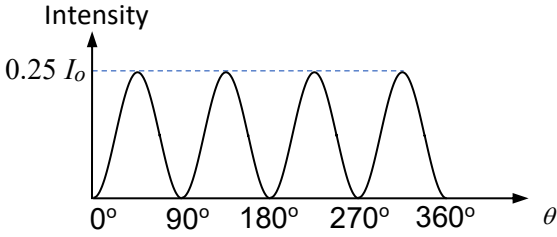


3	(a)	The direction of vibration of <i>electric field</i> is restricted to one direction in a plane that is normal to the direction of energy transfer (propagation).		[1]								
	(b)	(i)	When polariser X is oriented such that its direction of polarisation is not perpendicular to A, a component of the polarised light along the direction of polarization of X passes through X. The light component passing through X has a component along the direction of polarization of A that can pass through A. Thus the intensity of the transmitted light is not zero.	[1] [1]								
			1. <table><thead><tr><th>θ</th><th>Intensity of light transmitted by polariser A (in terms of I_o)</th></tr></thead><tbody><tr><td>0°</td><td>0</td></tr><tr><td>45°</td><td> $(\cos^2 45^\circ)^2 I_o = I_o / 4 = 0.25 I_o$</td></tr><tr><td>$60^\circ$</td><td>$(\cos^2 60^\circ)(\cos^2 30^\circ) I_o = 3 I_o / 16 = 0.1875 I_o$</td></tr></tbody></table>	θ	Intensity of light transmitted by polariser A (in terms of I_o)	0°	0	45°	 $(\cos^2 45^\circ)^2 I_o = I_o / 4 = 0.25 I_o$	60°	$(\cos^2 60^\circ)(\cos^2 30^\circ) I_o = 3 I_o / 16 = 0.1875 I_o$	[1] [1]
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			2. - min intensities are correct at $0^\circ, 90^\circ, 180^\circ, 270^\circ$ and 360°. Max intensities ($0.25 I_o$) at $45^\circ, (90^\circ + 45^\circ), (180^\circ + 45^\circ)$, and $(270^\circ + 45^\circ)$ - correct shape 	[1] [1]								
	(c)	(i)	Let P_s = Power emitted by source uniformly in all directions I_{78} = Intensity at 78.0 m away I_{300} = Intensity at 300 m away Power = Intensity x area $P_s = (I_{78})(4\pi \times 78.0^2) = (I_{300})(4\pi \times 300.0^2)$ $I_{300} = (0.026) \left(\frac{4\pi \times 78.0^2}{4\pi \times 300.0^2} \right) = 0.00176 \text{ W m}^{-2}$	[1] [1]								
		(ii)	Let P_r = power received by the microphone diaphragm $P_r = (I_{300})(\text{area of microphone})$ $P_r = (0.00176)(3.2 \times 10^{-4}) = 5.63 \times 10^{-7} \text{ W}$	[1]								